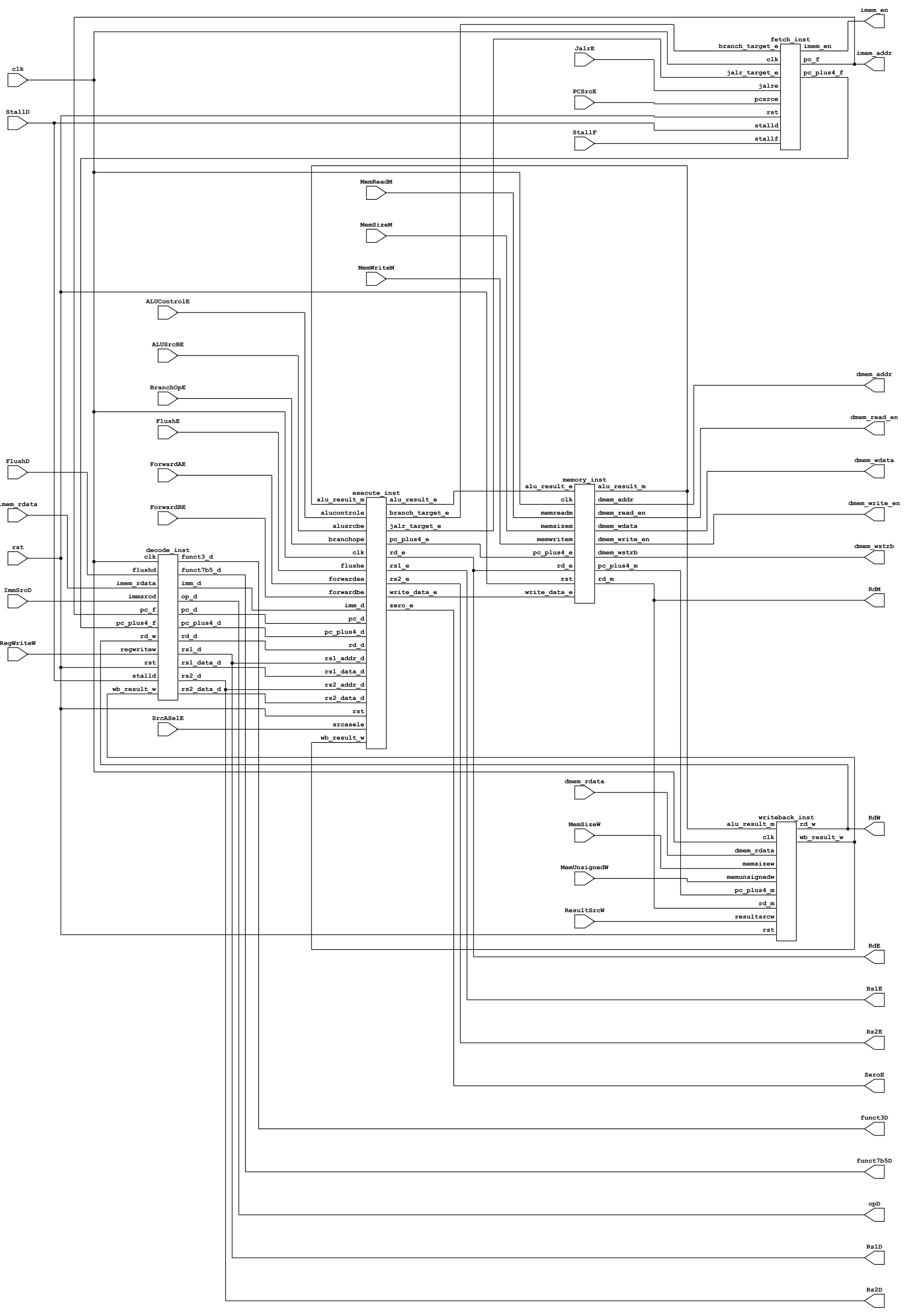
clk
StallD
FlushD
imem_rdata
rst
ImmSrcD
RegWriteW
ALUControlE
ALUSrcBE
BranchOpE
FlushE
ForwardAE
ForwardBE
SrcASelE
MemReadM
MemSizeM
MemWriteM
imem_en
JalrE
PCSrcE
StallF
fetch_inst
branch_target_e
clk
jalr_target_e
jalre
pcsrce
rst
stalld
stallf
imem_en
imem_addr
pc_f
pc_plus4_f
decode_inst
clk
flushd
imem_rdata
immsrcd
pc_f
pc_plus4_f
rd_w
regwritew
rst
stalld
wb_result_w
funct3_d
funct7b5_d
imm_d
op_d
pc_d
pc_plus4_d
rd_d
rs1_d
rs1_data_d
rs2_d
rs2_data_d
execute_inst
alu_result_m
alucontrole
alusrcbe
branchope
clk
flushe
forwardae
forwardbe
imm_d
pc_d
pc_plus4_d
rd_d
rs1_addr_d
rs1_data_d
rs2_addr_d
rs2_data_d
rst
srcasele
wb_result_w
alu_result_e
branch_target_e
jalr_target_e
pc_plus4_e
rd_e
rs1_e
rs2_e
write_data_e
zero_e
memory_inst
alu_result_e
clk
memreadm
memsizem
memwritem
pc_plus4_e
rd_e
rst
write_data_e
alu_result_m
dmem_addr
dmem_read_en
dmem_wdata
dmem_write_en
dmem_wstrb
pc_plus4_m
rd_m
dmem_addr
dmem_read_en
dmem_wdata
dmem_write_en
dmem_wstrb
RdM
dmem_rdata
MemSizeW
MemUnsignedW
ResultSrcW
writeback_inst
alu_result_m
clk
dmem_rdata
memsizew
memunsignedw
pc_plus4_m
rd_m
resultsrcw
rst
rd_w
wb_result_w
RdW
RdE
Rs1E
Rs2E
ZeroE
funct3D
funct7b5D
opD
Rs1D
Rs2D